

## A Probabilistic and Information-Theoretic View of AI Agents

*How an agent decides what to believe,  
what to ask, and when to act*

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**What this document is.** This is the Week 2 project brief. It extends the Week 1 project. You keep the same problem. You keep the same agent. You add beliefs, information value, cost, and a decision policy.

**What you submit.** One PDF preprint, written by you, in IJCAI-style format. Code and notebooks are optional and go beside the PDF.

**When you submit.** There is no deadline. Read [Section 3](#) before you plan your time.

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## 1 How to read this document

This document is long. That is on purpose. It is a reference, not a checklist you race through.

Read it in this order.

1. Read Sections 3 to 6 today. They take fifteen minutes. They tell you what the project is and how much time to give it.
2. Read Section 7 slowly. It teaches every idea you need, with numbers you can check by hand.
3. Then work through Sections 10 to 19. These are the actual build.
4. Write the paper using Section 21 as the outline.
5. Use Section 25 whenever you are stuck or unsure what is being asked.

If after reading Sections 1 to 3 you still do not know where to start — which is a completely normal reaction to a document this size — go straight to Section 2. There is a tool that will work that out with you.

Two notes on style.

Every idea in this document is explained twice. Once in plain words. Once with a small worked example using real numbers. If the plain words are enough for you, skip the numbers. If the words feel vague, do the numbers by hand. The numbers are small on purpose.

You are not being graded on mathematics. You are being graded on whether you can reason about an uncertain world, design something, test it, watch it fail, explain the failure, and change your design.

## 2 Two helpers: the readiness interview and the coach

This brief is long, and it is written for a room full of people at very different starting points. Some of you have shipped models. Some of you saw the word “entropy” for the first time last week. A single document cannot give both of you the right next step.

So the repository ships two **skill files** you can install into Claude Code. Neither is required. Both exist because the honest answer to “what should I do first?” depends entirely on who is asking.

### 2.1 Skill 1 — the readiness interview (run this first)

week2-readiness does not teach you anything. It **interviews you**, then tells you what you personally need.

It asks about your Week 1 problem, what you already built, which concepts you can actually explain versus which ones you have only heard, whether you can write code, how much time you have per week, and where you are quietly stuck. Then it produces **your** roadmap: what to gather before you start, what to learn first and in what order, which sections of this brief matter most for you, which ones you can skim, and roughly how many sittings the whole thing will take at your pace.

#### “I don’t know” is the most useful answer you can give it

The interview will ask you things you cannot answer. That is the design.

Say “I don’t know” and mean it. Say “I’ve heard of KL divergence but I couldn’t explain it.” Say “I don’t actually understand what my hidden states are.” Every one of those answers makes the roadmap sharper.

If you bluff — and people do, out of embarrassment, to a piece of software — you will get a roadmap built for somebody else, skipping exactly the things you needed. Nobody sees your answers. There is nothing to defend.

## 2.2 Skill 2 — the coach (run this while you build)

week2-coach walks beside you through the build, one stage at a time: hidden states, likelihoods, belief updates, information value, cost, thresholds, the experiment, failure analysis, the paper. Each stage has a gate — an explicit check on whether that piece is actually solid before you move on.

It will explain any concept in this document as many times and as many ways as you need. It will check your arithmetic, debug your code, and tell you honestly which part of your draft is weakest.

**It will not write your project.** If you ask it to, it will say no once, briefly, and then offer to do the hardest useful thing it legitimately can. That refusal is deliberate. The course evaluates whether you can reason under uncertainty, and a paper you did not think through is a paper you cannot defend in the interview it was meant to prepare you for.

## 2.3 Installing them

```
mkdir -p ~/.claude/skills/week2-readiness ~/.claude/skills/week2-coach
cp week2/skill/readiness/SKILL.md ~/.claude/skills/week2-readiness/SKILL.md
cp week2/skill/coach/SKILL.md      ~/.claude/skills/week2-coach/SKILL.md
```

Then run `/week2-readiness` and answer honestly. Later, run `/week2-coach` and tell it which stage you are on.

If you use ChatGPT or Gemini instead of Claude Code, the skill files are plain text. Open one, paste its contents in as your first message, and it works the same way.

## 3 Pacing: this is not a seven-day sprint

Read this section carefully. It is the part students most often get wrong.

### The pacing rule

“Week 2” is the name of the deliverable. It is not a deadline.

If you are full-time on this course, you may finish in a week. If you can give two hours a day, you will finish comfortably over two to three weeks. If you are a working professional and you submit this deliverable in Week 5 or Week 6, that is completely acceptable. Nobody will penalise you for it.

We do not enforce deadlines on deliverables. We have consistently seen that the students who give a deliverable the right amount of time produce better work than the students who rush to match a calendar.

What this means in practice:

- Do not submit a thin paper on day seven to be on time. There is no “on time”.
- Do not skip the experiment because the week ended. The experiment is the project.
- Do keep moving. “No deadline” is not “no progress”. Aim for one section of real work per sitting.
- Do submit when the work is honest and complete, whenever that is.

A reasonable plan for someone with two hours a day:

| Sitting    | What you do                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------------------|
| 1–2        | Re-read your Week 1 paper. Write the hidden states and priors again. Build the belief-state table (Section 11). |
| 3–4        | Learn the concepts in Section 7 by working the small examples on paper.                                         |
| 5–6        | Compare three evidence sources. Build the question table (Sections 12, 13).                                     |
| 7–8        | Define costs, thresholds, escalation, and the stop rule (Section 15).                                           |
| 9–12       | Build and run the experiment. Two policies minimum (Section 18).                                                |
| 13–14      | Read five failures properly. Classify them. Change one design decision because of them (Section 19).            |
| 15–18      | Write the paper. Make the figure and the tables.                                                                |
| Throughout | The five LinkedIn posts, the Reddit and X discussions (Section 20). Do these as you go, not at the end.         |

## 4 What changed from Week 1

In Week 1 you built an agent that had to decide something when the truth was not visible.

You started here:

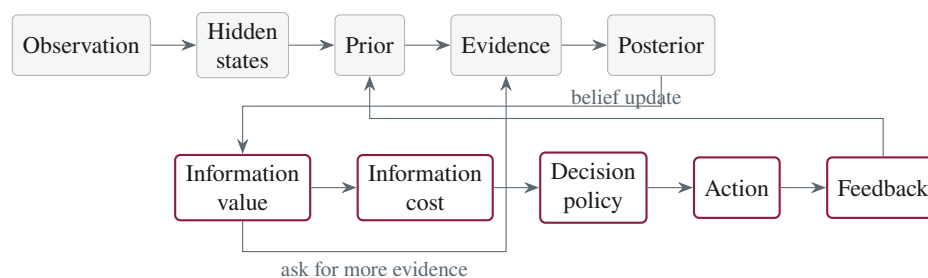
Input → Agent → Decision

and moved to here:

Input → Hidden states → Beliefs → Evidence → Posterior → Action

Week 2 adds three things that Week 1 did not have: the agent now *chooses* what evidence to look at, it *pays* for that evidence, and it *learns* from what happened after it acted.

Problem → Hidden states → Beliefs → Evidence →  
Information value → Decision → Cost → Action → Feedback → Belief update



**Figure 1:** Week 1 is the top row. Week 2 adds the bottom row and the two loops. The dark boxes are new.

The point is not to make the architecture complicated. The point is that you should be able to say, for each new box, *why it exists* and *what breaks without it*.

## 5 Continue the same problem

Use the same problem you chose in Week 1.

Do not start a new project. Do not choose a new problem. Do not throw away Week 1.

If you genuinely cannot continue your Week 1 problem, say so in your paper and explain why. That explanation is itself useful research. Do not silently switch.

Throughout this document we use one running example so that every idea has something concrete attached to it:

The agent receives an email from a known supplier asking to change the bank account for future payments. The agent must **pay**, **verify**, or **hold** the request, because the true identity and intent of the sender are not directly observable.

If your problem is different, the structure still applies. Replace “supplier”, “bank change”, and “pay” with your own words. Every table and every calculation in this document has the same shape for any problem where something important is hidden.

## 6 Your central research question

Your whole Week 2 project answers one question:

How can an AI agent use probabilistic reasoning and information theory to decide *what it should believe, what information it should seek, and when it should act*?

You may narrow this question to fit your problem. In fact you should. A narrowed version of the running example:

*When a supplier requests a bank-account change, which verification step should the agent perform next, and at what belief should it stop verifying and either pay or hold?*

Write your narrowed question down before you build anything. Put it in the introduction of your paper. Every section of the paper should visibly serve it.

## 7 The concepts you need

This section is deliberately not a textbook.

For each concept you get three things: what it means in plain words, a concrete example, and a **prompt you can paste into an AI assistant** to learn it properly. You will not find the equations here. That is on purpose.

Two reasons for that choice. First, a formula you cannot read teaches you nothing, and most of you are beginners — reading  $-\sum p \log p$  before you understand what it is for is how people decide they are bad at mathematics. Second, and more importantly, **going and learning the thing yourself is the actual assignment**. A brief that hands you every definition trains you to wait for definitions. Use the prompts. Argue with the answers. Come back with the version you can explain.

Every concept below carries the same mark:

[extend this yourself]

Some of these you have met in a session and some you have not, and it does not matter which. The instruction is identical either way. Whatever you were given, go past it — the sessions are a starting point, never the boundary.

### 7.1 How to use the prompts

Paste the prompt. Read the answer. Then do the part people skip: **close the chat and write the explanation in your own words from memory**. If you cannot, you have not learned it. Go back and ask the assistant to explain it differently.

Every prompt below ends with a line asking the assistant to connect the concept to your problem. Replace [YOUR PROBLEM] with your own one-sentence problem statement each time. That last line is what turns a definition into something you can use.

### Two things the assistant will get wrong

**It will over-formalise.** If it answers with notation, reply: “That was too mathematical. Explain it again with no symbols at all, using an example a ten-year-old could follow.” Keep asking until it lands. This is a normal part of using these tools well.

**It will agree with you.** If you say “so entropy is basically accuracy, right?” it may well say yes. Test it the other way: “Give me the strongest reason what I just said is wrong.”

## 7.2 The setup we will reuse

Every example in this document uses one running case, so you can watch the same four hidden states move through all of the concepts without switching contexts.

| Hidden state  | Prior | Meaning                                                |
|---------------|-------|--------------------------------------------------------|
| Genuine       | 0.40  | The supplier really did change banks.                  |
| Compromised   | 0.20  | The real supplier’s mailbox is under attacker control. |
| Impersonation | 0.25  | Someone is imitating the supplier from outside.        |
| Malicious     | 0.15  | A deliberate fraud attempt, possibly an insider.       |

These four numbers add to 1.00. That is not decoration. It is what makes them a belief. If they do not add to one, the agent is claiming something is possible that is not on its list, or claiming the same possibility twice.

### Where do these four numbers come from?

Yours will come from one of three places, and you must say which in your paper:

1. **Data.** You counted. “Of 1,200 bank-change requests last year, 480 were genuine.”
2. **Domain judgement.** You asked someone who does this job. Say who, and when.
3. **Assumption.** You made it up as a starting point. This is allowed. Label it clearly as hypothetical.

Inventing numbers and presenting them as measured is the one thing that will fail your submission. Inventing numbers and labelling them as assumptions is normal research practice.

## 7.3 Probability and belief

[extend this yourself]

A probability is a number between 0 and 1 saying how strongly you expect something. It is not a fact about the world. It is a statement about *your knowledge* of the world.

Two careful people can hold different probabilities for the same event because they have seen different things. When new evidence arrives both should move, and if they see enough of the same evidence they should end up close together.

**Why your agent cares.** An agent that outputs one answer has thrown away everything it was unsure about. An agent that holds a belief across several possibilities can tell you it is unsure — and being able to say “I don’t know” is what makes escalation to a human possible.

**Prompt — paste this into ChatGPT, Claude, or Gemini**

I am a beginner. Explain the difference between a probability, a prior, a posterior, and a likelihood, with no equations at all. Use one everyday example and keep the same example for all four. Then tell me the most common mistake beginners make in confusing likelihood and posterior, and give me a test I can use to catch myself making it. Finally connect all four ideas to this problem: [YOUR PROBLEM].

**7.4 Updating a belief when evidence arrives****[extend this yourself]**

The mechanics are four columns of arithmetic. You should do this by hand at least once, because seeing it is worth more than being told it.

Say the agent runs an email authentication check (SPF). First it needs to know how often that check fails *under each hidden state*:

| Hidden state  | SPF fails | SPF passes |
|---------------|-----------|------------|
| Genuine       | 0.02      | 0.98       |
| Compromised   | 0.10      | 0.90       |
| Impersonation | 0.70      | 0.30       |
| Malicious     | 0.60      | 0.40       |

Read a row as: “if the message really is from an impersonator, SPF fails 70% of the time.” That is a **likelihood**. It runs in the opposite direction to what you actually want to know, and getting the direction backwards is the single most common error in this project.

Now suppose SPF fails. Multiply, add, divide:

| State         | Prior | Likelihood | Prior × Lik. | Posterior |
|---------------|-------|------------|--------------|-----------|
| Genuine       | 0.40  | 0.02       | 0.008        | 0.027     |
| Compromised   | 0.20  | 0.10       | 0.020        | 0.068     |
| Impersonation | 0.25  | 0.70       | 0.175        | 0.597     |
| Malicious     | 0.15  | 0.60       | 0.090        | 0.307     |
| <b>Total</b>  | 1.00  | —          | 0.293        | 1.000     |

Multiply each prior by its likelihood. Add the column. Divide each row by that total. That is Bayes’ theorem, and there is nothing else in it. Belief in impersonation went from 0.25 to 0.60.

**Prompt — paste this into ChatGPT, Claude, or Gemini**

I am a beginner. Walk me through Bayes’ theorem using only a table of numbers and no algebra. Use a medical test example first, then repeat the identical table structure for this problem: [YOUR PROBLEM]. Show every arithmetic step. Then tell me what goes wrong if I forget to divide by the total at the end, and what it means if my likelihoods are the same for every hidden state.

**7.5 Entropy****[extend this yourself]**

Entropy is **one number that says how confused you are**.

Zero means certain. Higher means your belief is spread out across many possibilities. The unit is **bits**, and one bit is roughly “one good yes/no question’s worth of uncertainty”.

**The example that makes it click.** You are thinking of an animal and I have to guess it. If there are eight equally likely animals, I need three yes/no questions: halve, halve, halve. Eight possibilities is three bits. If you tell me it is almost certainly a dog, we are down to nearly zero bits — I barely need to ask anything.

For the four hidden states above, entropy is about 1.90 bits. Four equal states would be exactly 2 bits. So the agent's prior is barely better than knowing nothing at all.

**Why your agent cares.** It converts “how sure am I?” into a single number you can track, compare, and plot. Watching entropy fall as evidence arrives is the clearest picture of your agent thinking that you will produce all week.

#### Prompt — paste this into ChatGPT, Claude, or Gemini

I am a beginner and I have never studied information theory. Explain entropy to me with no formula. Start with the guessing-game intuition, then tell me why the unit is called a bit. Give me three small examples: a fair coin, a coin that lands heads 99% of the time, and four equally likely options. Tell me what entropy is NOT (it is not accuracy, it is not error). Then show me how to compute it once, by hand, for these four numbers: 0.40, 0.20, 0.25, 0.15 --- and only then show me the formula, explaining what each piece of it is doing in words. Finally, connect it to this problem: [YOUR PROBLEM].

## 7.6 Information gain

[extend this yourself]

Information gain is **how much a question is worth**: how much your confusion drops because you asked it.

Before the SPF check, the agent sat at 1.90 bits of confusion. If the check comes back *failed*, the belief becomes the posterior column above and confusion drops to 1.37 bits — a drop of 0.53 bits.

**But here is the part almost everyone gets wrong.** Before you run the check you do not know whether it will fail. It fails about 29% of the time and passes about 71% of the time, and when it passes it tells you much less (confusion only falls to 1.62 bits). The honest value of *running* the check is the average across both outcomes, weighted by how likely each outcome is. That average works out to about **0.36 bits**, not 0.53.

#### The distinction that decides whether your paper is right

**0.53 bits** is what you learned *after* seeing a specific result.

**0.36 bits** is what the check is worth *before* you run it — and running it is the decision you actually have to make.

The second number is called **expected information gain**. Students routinely compute the first, call it the second, and rank their evidence sources in the wrong order because of it. If you take one piece of arithmetic seriously this week, take this one.

#### Prompt — paste this into ChatGPT, Claude, or Gemini

I am a beginner. Explain information gain with no equations. Use the game of twenty questions. Then explain the difference between (a) how much I learn after I see a specific answer and (b) how much I expect to learn before I ask the question, and why (b) is the one I should use to choose which question to ask. Give me a worked example with small numbers where (a) and (b) are noticeably

different. Then explain why “is it an animal?” is often a better question than “is it a dog?” even though both are yes/no. Connect it to this problem: [YOUR PROBLEM].

## 7.7 Conditional entropy

[extend this yourself]

Conditional entropy is **how confused you expect to still be after you get an answer.**

That is the 1.55 bits hiding inside the paragraph above — the average of 1.37 (if SPF fails) and 1.62 (if it passes), weighted by how often each happens. Information gain is just the drop from where you started to there.

**Why your agent cares.** It is the honest answer to “how much doubt will remain if I run this check?” Some checks leave you in almost exactly the same fog you started in. Conditional entropy is how you find those before you pay for them.

### Prompt — paste this into ChatGPT, Claude, or Gemini

Explain conditional entropy to someone who understands entropy but has never seen the conditional version, using no formulas. Make clear it is an average over all the answers I might get, not the entropy after one particular answer. Give me a small worked example with two possible answers where I can check the arithmetic by hand. Then tell me the relationship between conditional entropy and information gain in one sentence. Connect it to this problem: [YOUR PROBLEM].

## 7.8 Mutual information

[extend this yourself]

Mutual information asks: **how much does knowing one thing tell me about another thing?**

If it is zero, the two are unrelated and one is useless for predicting the other. The higher it is, the more one reveals about the other.

Here is the small surprise worth putting in your paper: **mutual information and expected information gain are the same quantity.** Two communities named the same idea differently — decision trees say “information gain”, communication theory says “mutual information”. When you notice that two things you learned separately are one thing, you have understood both.

### Mutual information catches what correlation misses

Take a number that can be  $-2$ ,  $-1$ ,  $1$ , or  $2$ , each equally likely. Now square it, giving  $4$ ,  $1$ ,  $1$ , or  $4$ .

**Correlation between the two is exactly zero.** The relationship is perfectly symmetric so it cancels itself out, and any correlation-based tool will report “no relationship here”.

**But knowing the first number tells you the second one perfectly.** Mutual information is 1 bit — the maximum possible.

Correlation asks “do these move together in a straight line?” Mutual information asks “does knowing one reduce my uncertainty about the other *in any way at all*?” For agent design the second question is the one you want, because your most useful evidence is very often not linearly related to your hidden state.

### Prompt — paste this into ChatGPT, Claude, or Gemini

Explain mutual information to a beginner with no formulas. Tell me intuitively what it measures and what it means when it is zero. Then explain how it differs from correlation, using the example of a number and its square, and give me one more example from everyday life where two things are strongly related but correlation would report nothing. Then tell me why mutual information is symmetric and what that symmetry means practically. Finally, tell me how I could use it to decide whether an evidence source is worth collecting for this problem: [YOUR PROBLEM].

## 7.9 Cross-entropy

[extend this yourself]

Cross-entropy is **a score for how surprised your model is by reality**, on average. Lower is better. It is lowest when your model's beliefs match what actually happens.

**The property that matters.** It punishes being *confidently wrong* far more harshly than being unsure. Suppose the truth turns out to be A. If your model gave A a 50% chance, it pays a small penalty. If your model gave A a 1% chance, it pays a penalty roughly seven times larger. Confidence is only cheap when you are right.

This is why almost every classifier you will ever train is trained on cross-entropy, and it is also the behaviour you want in your agent: an agent that says “99% safe” about a fraudulent payment should be penalised much harder than one that shrugged and said “50/50”.

### Prompt — paste this into ChatGPT, Claude, or Gemini

Explain cross-entropy to a beginner with no equations. Frame it as “how surprised is my model, on average, by what really happens”. Show me with three small examples why being confidently wrong is punished much harder than being uncertain. Then explain why this is the loss function used to train classifiers, and what would go wrong if we used plain accuracy instead. Connect it to why a confident, badly calibrated AI agent is dangerous in this problem: [YOUR PROBLEM].

## 7.10 KL divergence

[extend this yourself]

KL divergence measures **how far apart two beliefs are** — specifically, how much you lose by believing the wrong one. It is zero when the two are identical.

You will hear it called a “distance”. It is not one, and the reason is worth understanding.

### The example that shows why it is not a distance

Reality: a fair coin. Heads half the time, tails half the time.

Your model: “heads, always. Tails is impossible.”

**Measured one way**, the disagreement is *infinite*. Half the time reality produces tails — an event your model called impossible — and being surprised by an impossible event is infinitely costly.

**Measured the other way round**, the disagreement is small and finite.

Same two beliefs, two different answers depending on which one you put first. A real distance cannot do that; the distance from Delhi to Mumbai does not depend on which city you name first. So KL divergence is called a *divergence*.

**The lesson for your agent, and it is a serious one:** assigning zero probability to something that

can actually happen is the most dangerous modelling error available to you. If “insider fraud” is not on your hidden-state list, no amount of evidence will ever make your agent consider insider fraud. It is not unlikely in your model. It is *impossible* in your model, and your agent will be infinitely surprised on the day it happens.

This is why Section 10 tells you to include a residual “something else” state.

#### Prompt — paste this into ChatGPT, Claude, or Gemini

Explain KL divergence to a beginner with no formulas. Start with what it is for: comparing two probability distributions, one of which is usually reality and one of which is usually my model. Explain what it means when it is zero. Then explain clearly why it is asymmetric, and construct a simple two-outcome example where measuring it one way gives a much bigger number than the other way. Explain why assigning zero probability to a possible event makes it infinite, and why that is a warning to anyone designing an AI agent. Only after all of that, show me the formula and explain what each piece is doing. Connect it to this problem: [YOUR PROBLEM].

### 7.11 Jensen–Shannon divergence

[extend this yourself]

Once you know KL divergence is asymmetric and can blow up to infinity, the obvious next question is: *is there a fixed version?* There is. That question — and following it — is exactly the behaviour this course is trying to build in you.

Jensen–Shannon divergence compares two beliefs by measuring each one against their average instead of against each other. Three things follow, and all three are practical:

- **It is symmetric.** Order does not matter any more.
- **It never goes to infinity**, even when one belief says something is impossible.
- **It is bounded**, so the number means something on its own rather than only in comparison.

**Why your agent cares, concretely.** Take the pattern of evidence your agent saw last quarter and the pattern it is seeing this month. Compare them. If the number jumps, the world has moved and your priors are now describing a world that no longer exists. That is a **distribution shift** alarm, built out of one idea from information theory — and KL divergence would have been the natural first choice and would have returned infinity the first time a genuinely new kind of evidence showed up.

#### Prompt — paste this into ChatGPT, Claude, or Gemini

I understand that KL divergence is asymmetric and can be infinite. Explain Jensen–Shannon divergence to me as the fix for those two problems, with no formulas at first. Explain the idea of comparing both distributions to their average and why that repairs both problems. Tell me what it means that it is bounded, and what its square root gives me. Then give me a practical example of using it to detect that the world has changed since a model was built. Finally show me the formula and explain each piece. Connect it to this problem: [YOUR PROBLEM].

## 7.12 Calibration

[extend this yourself]

An agent is **calibrated** if, across all the cases where it said “80% likely”, the thing actually happened about 80% of the time.

**This is not the same as accuracy**, and the difference matters enormously. An agent can be 95% accurate and badly calibrated: it says “99% certain” about everything, is right 95% of the time, and is therefore overconfident on every single case it ever sees.

### Why a confident agent can be more dangerous than an uncertain one

An uncertain agent escalates to a human. Its uncertainty is a signal that switches on a safety mechanism.

An overconfident agent never escalates. It is wrong at exactly the same rate — it has simply removed the one control that would have caught its mistakes.

Confidence is not a virtue in an agent. Calibrated confidence is. And every downstream thing you build this week — your threshold, your expected-cost calculation, your stop rule — takes the agent’s probabilities at face value. If those numbers are dishonest, everything built on them is dishonest too.

**How to check it.** Bucket your agent’s predictions (0–10%, 10–20%, and so on). For each bucket, compare the average confidence it stated against how often it was actually right. A calibrated agent’s points sit on the diagonal. This plot has a name and there is a standard summary number for the gap — find both.

### Prompt — paste this into ChatGPT, Claude, or Gemini

Explain calibration for a beginner, with no formulas. Make the difference between accuracy and calibration very clear with an example of a model that is highly accurate and badly calibrated, and another that is less accurate but well calibrated. Tell me which one I would rather deploy in a system that escalates uncertain cases to a human, and why. Then tell me the name of the standard plot for checking calibration and the standard number that summarises the error, and explain how to build both from a table of predictions and outcomes. Explain why large language models are often poorly calibrated. Connect this to this problem: [YOUR PROBLEM].

## 7.13 Expected value and expected cost

[extend this yourself]

Expected value is what you get on average if you make the same choice many times: each outcome weighted by how likely it is. Expected cost is the same idea, counting losses instead.

The rule your agent lives by follows from this, and it is not obvious:

The agent should **not** choose the most likely state. The agent should choose the **action with the lowest expected cost**.

Those are frequently different actions. If a fraudulent payment costs four hundred times more than annoying a supplier, the agent should hold the payment even when it is fairly confident the request is genuine. “Most likely” and “best” are simply not the same question, and Section 15 works out exactly where they part company.

**Prompt — paste this into ChatGPT, Claude, or Gemini**

Explain expected value and expected cost to a beginner with no formulas, using a bet and then an insurance decision. Then explain, with a concrete example, why the action with the highest probability of being right is sometimes not the action you should take. Make the example one where the two errors have very different costs. Finally explain what ``opportunity cost'' means in that same example. Connect it to this problem: [YOUR PROBLEM].

**7.14 Value of information****[extend this yourself]**

This is the most important idea in Week 2, so read this subsection twice.

Value of information is **how much better off you are for having asked**, minus what asking cost you.

Notice what it is *not* measured in. It is not measured in bits. It is measured in whatever your outcomes are measured in — rupees, hours, lives, reputation, delay.

Information has value only when some possible answer would make you **do something different**.

If every answer leads to the same action, the value of that information is zero, minus its cost. Do not buy it — no matter how many bits it carries.

That rule is the whole of Section 14, and it is where most students' thinking changes this week.

**Prompt — paste this into ChatGPT, Claude, or Gemini**

Explain ``value of information'' to a beginner with no formulas. Make the key point very clearly: information is only worth something if it could change what I decide to do. Give me one example where an extremely informative test is worthless because it cannot change the decision, and one example where a cheap, much less informative test is worth buying. Then explain how to decide whether to pay for a test, in plain steps I can follow. Explain when I should stop gathering information entirely. Connect it to this problem: [YOUR PROBLEM].

**7.15 Exploration and exploitation****[extend this yourself]**

**Exploitation** is acting on what you already believe. **Exploration** is paying something now to learn something that improves your later decisions.

Your agent faces this every time it considers a verification step. Skipping the phone call is exploitation. Making the call is exploration. A pure exploiter is fast and eventually wrong in a way it never finds out about. A pure explorer never gets anything done.

Your stop rule, in Section 15, is your answer to this trade-off. Present it as one.

**Prompt — paste this into ChatGPT, Claude, or Gemini**

Explain exploration versus exploitation to a beginner with no formulas or algorithms. Use an everyday example like choosing a restaurant. Then explain what goes wrong at each extreme: an agent that only exploits, and an agent that only explores. Tell me the common simple strategies people use to balance the two and what each one assumes. Connect it to an agent that must decide whether to run one more verification check before acting on this problem: [YOUR PROBLEM].

## 7.16 Distribution shift

[extend this yourself]

Your agent's priors were set from how the world used to be. Distribution shift is what happens when the world stops being that way and nobody tells the agent.

The dangerous part is that nothing breaks loudly. The agent keeps producing confident answers from beliefs that no longer describe anything real. Fraud tactics change. A supplier gets acquired. Your company enters a new market. Your priors quietly become fiction.

**Why your agent cares.** Detecting this is one of the few genuinely open problems you can touch this week. The Jensen–Shannon alarm described above is one workable answer. There are others. Finding a second one and comparing them would make an excellent contribution.

### Prompt — paste this into ChatGPT, Claude, or Gemini

Explain distribution shift to a beginner with no formulas. Distinguish clearly between the input data changing and the underlying relationship changing, and give an example of each. Then explain why a model can keep producing confident answers while being completely wrong, and why nobody notices for a long time. Give me three practical ways a deployed system could detect that it has happened, and the weakness of each. Finally, tell me what a system should actually do once it suspects the world has shifted. Connect it to this problem: [YOUR PROBLEM].

## 7.17 The whole set, on one page

| Concept                      | Why your agent needs it                                              |
|------------------------------|----------------------------------------------------------------------|
| Probability, belief          | Lets the agent hold several explanations instead of guessing one.    |
| Belief update (Bayes)        | Turns evidence into a changed mind.                                  |
| Entropy                      | One number for how confused the agent is right now.                  |
| Information gain             | How much a question is worth before you ask it.                      |
| Conditional entropy          | How much doubt will still remain after you ask.                      |
| Mutual information           | Whether an evidence source is related to what you care about at all. |
| Cross-entropy                | Scores a model, and punishes confident wrongness hardest.            |
| KL divergence                | Compares two beliefs; warns you about impossible-in-model events.    |
| Jensen–Shannon divergence    | The symmetric, finite version; a practical drift alarm.              |
| Calibration                  | Whether the agent's confidence can be trusted at all.                |
| Value of information         | Whether an answer is worth what it costs.                            |
| Exploration and exploitation | When to stop asking and act.                                         |
| Distribution shift           | Whether the agent's priors still describe the real world.            |

Thirteen concepts, one instruction for all of them: go further than you were given.

### These prompts teach you concepts. They do not do your project.

Everything above is a prompt to *understand something*. None of them asks an assistant to produce your hidden states, your priors, your costs, your results, or your paragraphs.

Keep that line where it is. A concept explained to you is yours once you can re-explain it. A result generated for you is never yours, and you will not survive the first question about it.

## 8 The extension rule

Your paper must contain concepts you went and found for yourself. Sessions and briefs are a starting point. Neither is a syllabus, and neither is a boundary.

We are not testing whether you can memorise a syllabus. We are testing whether, when you hit the edge of what you were told, you can push one step further on your own. This is the single most transferable skill in this course.

The behaviour looks like this. You take something you know, ask the obvious next question, and follow it:

| You were taught                               | The question that takes you further                                                                            |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Entropy measures uncertainty.                 | “What if I want to compare two beliefs, not measure one?” → KL divergence                                      |
| KL divergence compares distributions.         | “It gives different answers depending on the order. Is there a symmetric version?” → Jensen–Shannon divergence |
| Information gain tells me a question’s value. | “How much does observing $X$ tell me about $Y$ specifically?” → Mutual information                             |
| Mutual information measures dependence.       | “How is that different from correlation?” → Non-linear dependence                                              |
| The agent outputs a probability.              | “Is that number trustworthy?” → Calibration, reliability diagrams                                              |
| The agent has priors.                         | “What if the world changed since I set them?” → Distribution shift                                             |
| The agent picks the best question.            | “What if the best question is expensive?” → Value of information                                               |

Section 7 walks the first few steps of this ladder with you and gives you thirteen concepts to extend. Those thirteen are the floor, not the ceiling.

**Do not stop where this document stops.** Find at least two concepts that are not in this document at all. Learn them well enough to explain simply. Use them for something real. Name them in your paper and say how you found them.

### Where the next two concepts come from

Not from a list. From your own results.

The move is always the same: something in your project does not work, or works for a reason you cannot explain, and you go looking for the name of it. “My agent is 94% accurate and still loses money” sends you towards cost-sensitive learning. “My two evidence sources keep agreeing even when they are both wrong” sends you towards conditional independence. “I have almost no fraud cases in my data” sends you towards class imbalance.

A student who collects concepts from a syllabus produces a paper that lists them. A student who collects concepts from their own failures produces a paper that uses them. We can tell the difference immediately, and so can an interviewer.

The general-purpose prompt, when you have a name but no understanding:

Explain [CONCEPT] to me as if I have never studied machine learning. Give me the plain-English meaning first, in three sentences and no symbols. Then give me one worked example with numbers small enough for me to check by hand. Then tell me one situation where using this concept would be a mistake. Then connect it to this problem: [YOUR PROBLEM].

The last two lines matter most. A concept you cannot misuse is a concept you have not understood,

and a concept you cannot connect to your problem does not belong in your paper.

Then close the chat and write the explanation in your own words from memory. If you cannot, you have not learned it yet. Go back.

## 9 Reading papers (optional this week)

There is no mandatory reading list for Week 2.

If you want to read research papers, do. Choose beginner-friendly ones connected to your own problem. Good starting points are survey articles, blog explanations by the authors, and the introduction sections of papers — which are usually written for outsiders even when the rest is not.

If you do read something, do not summarise it in your paper. Connect it. For each paper or article, answer these, briefly:

1. What problem is it solving?
2. What is the simplest version of the idea?
3. What does it assume?
4. What information does the system use, and what decision does it make?
5. What is the main experiment, and what did the authors find?
6. What limitation did you notice?
7. **How does this change my agent's design?**

The last one is not optional. If a paper changed nothing in your design, it does not belong in your Related Work section.

### Two rules that do not bend

**Never cite something you did not read.** An AI assistant can produce a plausible citation list in two seconds. Some of those papers will not exist. Every reference in your bibliography must be something you opened.

**Never publish a claim you cannot explain.** If someone asks you at an interview what your paper meant on page 4 and you cannot answer, the paper is not yours.

It is perfectly fine to use an AI assistant to make a hard paper readable. Ask it to explain the paper in simple language, then verify the explanation against the actual paper. That is research. Pasting its summary into your Related Work section is not.

**Optional exercise, strongly recommended.** Take one paper or article and prepare a five-minute explanation for a friend who has never studied machine learning. Cover: the problem, the intuition, what the researchers actually did, what happened, what it fails to explain, and how it connects to your agent. If you can do this without looking at the PDF, you understand it.

## 10 Build the probabilistic view of your agent

Rebuild your Week 1 architecture with these twelve components. For each one, write a short paragraph about *your* problem. This becomes Section 5 of your paper.

| #  | Component         | The question it answers                                                                                                                                          |
|----|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Observation       | What does the agent actually receive? Be concrete. Not “an email” but “sender address, display name, SPF/DKIM result, message body, attachment, thread history”. |
| 2  | Hidden state      | What is genuinely unknown and matters? Usually: who this really is, or what they intend.                                                                         |
| 3  | Possible worlds   | What are all the plausible explanations? List every one, including boring ones. Include a residual “something else” state.                                       |
| 4  | Prior belief      | What did the agent believe before this evidence? Where did that number come from?                                                                                |
| 5  | Evidence          | What new signals can arrive or be requested?                                                                                                                     |
| 6  | Posterior belief  | How does belief change? Show one full Bayes calculation by hand in the paper.                                                                                    |
| 7  | Uncertainty       | How confused is the agent now? Report entropy in bits.                                                                                                           |
| 8  | Information value | Which available evidence would reduce that uncertainty most?                                                                                                     |
| 9  | Cost              | What does obtaining it cost in money, time, human attention, and relationship damage?                                                                            |
| 10 | Decision          | Which action minimises expected cost, given the belief and the costs?                                                                                            |
| 11 | Feedback          | What happens after acting, and when do you find out you were wrong?                                                                                              |
| 12 | Belief update     | How does the outcome change future priors?                                                                                                                       |

### Component 11 is the one people skip

Feedback delay decides how much your agent can learn. If a wrong “pay” decision is discovered in two days, you can update quickly. If it surfaces at the quarterly audit, your agent runs blind for three months and its priors slowly drift away from reality.

Worse, feedback is usually *asymmetric*. When the agent holds a payment, someone complains and you learn immediately that it was genuine. When the agent pays a fraudster, nobody complains. So the agent hears about its false positives loudly and its false negatives never. An agent that learns from this feedback will drift towards paying everything. Write about this. It is real, it affects almost every problem in this cohort, and it is exactly the kind of observation that makes a paper good.

## 11 The belief-state table

Build this table for your own problem. It is the spine of your paper.

| Hidden state  | Prior | Evidence observed     | Posterior | Implied action |
|---------------|-------|-----------------------|-----------|----------------|
| Genuine       | 0.40  | SPF fail, new account | 0.03      | Hold           |
| Compromised   | 0.20  | SPF fail, new account | 0.07      | Verify         |
| Impersonation | 0.25  | SPF fail, new account | 0.60      | Hold           |
| Malicious     | 0.15  | SPF fail, new account | 0.30      | Hold           |
| Entropy       |       | 1.90 bits →           | 1.37 bits |                |

*These numbers are illustrative and hypothetical. Yours must be your own.*

Rules for this table:

- Priors must sum to 1.00. Posteriors must sum to 1.00. Check the arithmetic.
- State where each prior came from — data, judgement, or assumption.
- Show at least one posterior worked out in full, the way Section 7 does. Not the final number only.
- Report entropy before and after. It makes the update visible as one number.
- If the numbers are hypothetical, write the word “hypothetical” next to them.

### On honesty with numbers

Hypothetical numbers, clearly labelled, are completely acceptable and normal at this stage.

Hypothetical numbers presented as measured results are academic fraud. There is no middle ground and no partial credit. If you did not run it, do not report it.

## 12 Information selection: which evidence next?

Your agent has several things it could check. It cannot check everything, and it should not.

Compare at least **three** evidence sources. For each one, answer all eight questions below. Here is a fully worked comparison from the running example, using the priors from Section 7.

### 12.1 Three candidate checks

**Check A — SPF/DKIM authentication result.** Already computed: expected information gain = 0.36 bits. Cost: effectively zero. Time: milliseconds.

**Check B — Has this supplier used this bank account before?** A lookup in the payments database.

| Hidden state  | $P(\text{new account} \mid S)$ | $P(\text{known account} \mid S)$ |
|---------------|--------------------------------|----------------------------------|
| Genuine       | 0.30                           | 0.70                             |
| Compromised   | 0.90                           | 0.10                             |
| Impersonation | 0.95                           | 0.05                             |
| Malicious     | 0.95                           | 0.05                             |

The account comes back *new* 68% of the time, leaving the belief at (0.177, 0.265, 0.349, 0.210) and confusion at 1.95 bits. It comes back *known* 32% of the time, leaving (0.875, 0.063, 0.039, 0.023) and confusion at 0.73 bits.

Average those two, weighted by how often each happens:  $0.68 \times 1.95$  plus  $0.32 \times 0.73$  is 1.56 bits of remaining confusion. Starting from 1.90, this check is worth **0.34 bits**.

**Check C — Phone the supplier on the number already on file** (not any number in the email) and ask them to confirm.

| Hidden state  | Supplier confirms | Does not confirm |
|---------------|-------------------|------------------|
| Genuine       | 0.95              | 0.05             |
| Compromised   | 0.30              | 0.70             |
| Impersonation | 0.05              | 0.95             |
| Malicious     | 0.05              | 0.95             |

The supplier confirms 46% of the time, leaving 0.85 bits of confusion. They do not confirm 54% of the time, leaving 1.71 bits.

Averaged the same way, 1.31 bits of confusion remain. Starting from 1.90, this check is worth **0.59 bits** — the most informative of the three.

## 12.2 The comparison table

| Evidence                | Exp. IG   | Cost       | Time   | Bits per Rs. 100 | If it is wrong                                                     |
|-------------------------|-----------|------------|--------|------------------|--------------------------------------------------------------------|
| A. SPF/DKIM             | 0.36 bits | ≈Rs. 0     | 200 ms | very large       | Forwarding and mailing lists break SPF. Genuine mail fails.        |
| B. Bank-account history | 0.34 bits | ≈Rs. 2     | 2 s    | 17.0             | Supplier really did switch banks. Legitimate change looks new.     |
| C. Phone callback       | 0.59 bits | Rs. 300    | 30 min | 0.20             | Wrong number on file, or the person who answers was also deceived. |
| D. Full forensic audit  | 0.90 bits | Rs. 10,000 | 2 days | 0.009            | Rarely wrong, but the request is stale by the time it finishes.    |

*Illustrative numbers. Cost for A and B is amortised infrastructure cost; cost for C is 30 minutes of analyst time.*

## 12.3 What this table already tells you

Three things worth writing about in your paper, all visible in the numbers above.

**First, the most informative check is the worst value.** The forensic audit carries the most bits and is close to useless in practice: 0.009 bits per hundred rupees, against 17 bits for the database lookup. Ranking evidence by information alone gives you exactly the wrong order.

**Second, cheap checks should run first, always.** A and B together cost about two rupees and two seconds. There is no situation where you would skip them. Order matters: run the free checks, then re-evaluate whether the expensive one is still needed. Often it is not, because the belief has already crossed a threshold.

**Third, and most interesting: information can increase uncertainty.** Look at Check B. Prior entropy is 1.90 bits. If the account comes back *new*, posterior entropy is **1.95 bits**. The agent knows more and is more confused than before.

This is not a bug and it is not an error in the arithmetic. Before the check, the agent leaned towards “genuine” at 0.40. Finding a new account knocks “genuine” down and leaves three remaining explanations at roughly similar strength — and a flat belief has high entropy. The agent has correctly moved from “probably fine” to “definitely something, but I do not know which something”.

### Reconciling this with the theory

*Expected* information gain is never negative:  $I(S; E) \geq 0$  always. On average, evidence helps. But the gain from any *single outcome* can absolutely be negative. Averaged over both outcomes, Check B still gives +0.34 bits.

This is a direct, worked answer to Advanced Question 4, taken from a real design rather than a textbook. If you find something like this in your own problem, that is exactly the kind of finding to build a LinkedIn post around.

### 13 Design five questions your agent could ask

Write five. Fill this table. Then explain which one the agent actually picks and why.

| Question                                                   | Possible answers                | Info value | Cost    | Time   | Would ask?                                     |
|------------------------------------------------------------|---------------------------------|------------|---------|--------|------------------------------------------------|
| Has this supplier used this account before?                | new / known                     | 0.34 bits  | Rs. 2   | 2 s    | Yes, always                                    |
| Did the message pass SPF and DKIM?                         | pass / fail / none              | 0.36 bits  | Rs. 0   | 200 ms | Yes, always                                    |
| Does the request match this supplier's historical pattern? | typical / unusual               | 0.21 bits  | Rs. 5   | 5 s    | Yes, always                                    |
| Can the supplier confirm on the number already on file?    | confirms / denies / unreachable | 0.59 bits  | Rs. 300 | 30 min | Only if still uncertain after the cheap checks |
| Was the request sent from a device or IP seen before?      | known / new                     | 0.28 bits  | Rs. 40  | 3 min  | Only if callback fails or is unreachable       |

*Illustrative numbers.*

Now write the reasoning explicitly. For the table above it reads:

*The agent always runs the three cheap checks first, in any order, because they cost under ten rupees combined and complete in seconds. It then recomputes its belief. If the posterior probability of a bad state is below 0.5% or above 60%, it acts immediately — the callback cannot change the decision, so its value of information is zero regardless of its bit count. It places the callback only in the middle band, where the extra 0.59 bits can actually flip the action. The device check is a fallback for when the supplier cannot be reached.*

Notice the structure of that paragraph: cheap first, recompute, then ask whether the expensive check could *change the decision*, and only spend if it could. That is the whole policy in four sentences. Yours should be similarly explicit.

### 14 Critical thinking challenge

**Is the question that gives the most information always the best question?**

This section is mandatory. Do not answer with a definition. Answer with numbers from your own problem.

Here is what a complete answer looks like, using the running example.

#### 14.1 The naive answer

Rank by information gain: the forensic audit wins with 0.90 bits, the callback is second with 0.59. Always run the audit.

#### 14.2 Why that is wrong

Bits are not the objective. Money, time, and correct decisions are the objective. Bits are only useful when they change what you do.

Take a request where the cheap checks have already left the agent at  $P(\text{bad}) = 0.75$ . The threshold for holding (derived in Section 15) is 0.0025. The agent is far above it. It will hold this payment.

Now ask the decisive question: *is there any outcome of the forensic audit that changes that action?* If the

audit comes back clean, the posterior drops — but not below 0.0025. The agent still holds. If it comes back dirty, the agent holds.

The audit carries 0.90 bits and changes nothing. It saves Rs. 0 and costs Rs. 10,000, so its value of information is minus ten thousand rupees. The most informative question available is the worst thing the agent could do.

### 14.3 The rule that follows

#### Information only has value when it can change an action

Before paying for any evidence, ask: *is there an outcome of this check that would make me do something different?*

If no, the value of that information is zero, minus its cost. Do not buy it. This holds no matter how many bits it carries.

If yes, compare the expected cost saved against the cost of obtaining it. Buy it only if the saving is larger.

### 14.4 The case where it does pay

Take a request sitting at  $P(\text{bad}) = 0.35$ , on a payment of Rs. 8,00,000.

- Hold now: the request is genuine with probability 0.65, so expected cost of a wrong hold is  $0.65 \times \text{Rs. } 2,000 = \text{Rs. } 1,300$ .
- Pay now: expected cost is  $0.35 \times \text{Rs. } 8,00,000 = \text{Rs. } 2,80,000$ .
- Callback for Rs. 300: it resolves most of the ambiguity. If the supplier confirms, the agent pays with far more confidence. If not, it holds with far more confidence. The expected cost after the callback drops to roughly Rs. 900.

Value of information  $\approx \text{Rs. } 1,300 - \text{Rs. } 900 - \text{Rs. } 300 = \text{Rs. } 100$ . Positive, so make the call. Marginally — and that is realistic. Most good decisions are not dramatic.

### 14.5 What to write in your paper

Give one case from your problem where the highest-information question is the right choice, and one case where it is not. Show the numbers. State the rule you extracted. Two paragraphs is enough if the numbers are real.

## 15 Decision-making under uncertainty

Your agent does not need certainty. It needs a rule for when it has enough.

### 15.1 Where thresholds actually come from

Most students write “if the probability of fraud exceeds 80%, hold the request.” Then the obvious question: **why 80%**? Why not 70%, or 90%?

The honest answer is that 80% came from nowhere. Thresholds are not chosen. They are *derived* from the cost of being wrong in each direction.

Two errors, two costs:

- **False positive:** hold a genuine request. It costs Rs. 2,000 — delay, supplier annoyance, some staff time.

- **False negative:** pay a fraudulent request. It costs Rs. 8,00,000 — the payment, mostly unrecoverable.

Now find the belief at which the two mistakes cost the same. Below it you should pay; above it you should hold. There is one line of arithmetic, and you can do it in your head:

**Break-even belief** = cost of a false positive  $\div$  (cost of a false positive + cost of a false negative)

$2,000 \div (2,000 + 8,00,000) = 0.0025$ , or one quarter of one percent.

### The threshold is 0.25%, not 80%

This result surprises almost everyone the first time. It follows directly from the costs. Paying a fraud is 400 times more expensive than annoying a supplier, so the agent should hold on a quarter of a percent of suspicion.

Answer the follow-up before your reader asks it: *if the threshold is really 0.25%, does the agent hold everything?* Nearly, if “pay” and “hold” are the only two actions. Which is why real systems have a third action.

The verify action costs Rs. 300 and resolves most of the doubt. With three actions the policy becomes: pay below  $p_{\text{low}}$ , verify between  $p_{\text{low}}$  and  $p_{\text{high}}$ , hold above  $p_{\text{high}}$ . The cheap middle action is what makes the system usable. This is the central design insight of the whole exercise, and it falls out of the arithmetic rather than out of intuition.

## 15.2 What you must answer about your threshold

1. What is your threshold, and what cost ratio produced it?
2. What happens at a threshold ten points lower? How many more false positives?
3. What happens ten points higher? How many more false negatives?
4. What is the cost of a false positive in your problem? Include the non-monetary parts.
5. What is the cost of a false negative? Same.
6. Should the threshold move with context — payment size, supplier age, time of month?
7. Which cases go to a human regardless of the probability?

Question 6 deserves real thought. A single fixed threshold implicitly claims that a Rs. 5,000 payment and a Rs. 50,00,000 payment deserve identical scrutiny. They do not, and your own arithmetic already knows this: the cost of a false negative grows with the payment, so the break-even belief should fall as the amount rises. A threshold that moves with the stakes is a stronger design than a constant, and it costs you one line of code.

## 15.3 The stop rule

When should the agent stop gathering and act? Any of these, whichever comes first:

1. **No remaining check can change the action.** Value of information is zero. Act.
2. **The best remaining check costs more than it saves.** Negative value of information. Act.
3. **Belief is outside the uncertain band.** Above  $p_{\text{high}}$  or below  $p_{\text{low}}$ . Act.
4. **Time ran out.** Some decisions expire. Act on the default safe action.

5. **Evidence stopped moving the belief.** Two checks in a row changed the posterior by less than some small amount. Further checks are probably measuring the same thing twice.

Rule 5 is worth investigating. Correlated evidence sources feel like more information and are not. Three sources that all key on the same underlying signal give you roughly one source's worth of bits while charging you three times.

### 15.4 Human escalation

Say explicitly when a human is involved. Reasonable triggers:

- The belief sits in the uncertain band after all affordable checks are exhausted.
- The cost of a wrong decision exceeds some ceiling, regardless of confidence.
- Two evidence sources disagree strongly.
- The case looks unlike anything in the historical data — a possible new hidden state.
- The agent's calibration has been drifting recently.

Note that the second trigger has nothing to do with probability. Some decisions should reach a human because of what is at stake, not because the agent is unsure. Confidence is not authorisation.

## 16 The umbrella problem

Apply the same reasoning to something small. This exercise is mandatory and takes twenty minutes.

There is a 35% chance of rain tomorrow. Should you carry an umbrella?

**Case A.** A normal walk. **Case B.** An important wedding, in clothes you cannot replace.

The probability is 35% in both. The answer is different. Explain why, then compute it.

### 16.1 Worked answer

Assign costs in “units of annoyance”. These are assumptions; state yours.

|                                 | Case A (walk)          | Case B (wedding)       |
|---------------------------------|------------------------|------------------------|
| Cost of carrying it and no rain | 2                      | 2                      |
| Cost of getting wet             | 5                      | 200                    |
| Average cost if you carry it    | 2.0                    | 2.0                    |
| Average cost if you do not      | $0.35 \times 5 = 1.75$ | $0.35 \times 200 = 70$ |
| <b>Decision</b>                 | Leave it               | Take it                |

Same 35%. Opposite decisions. The probability did not decide anything; the consequences did.

### 16.2 The break-even probability

There is a rain probability at which both choices cost the same. Above it, take the umbrella. Below it, leave it. It is one division:

**Break-even chance of rain** = cost of carrying it ÷ cost of getting wet

Case A:  $2 \div 5 = 0.40$ . A 40% chance of rain is the tipping point, and tomorrow is only 35%. Leave the umbrella.

Case B:  $2 \div 200 = 0.01$ . The tipping point is 1%, and tomorrow is 35% — thirty-five times over. Take it. You would take it at 2% as well.

### What this exercise is really for

It is the same structure as your agent, stripped of all domain complexity. Hidden state: rain or no rain. Belief: 0.35. Actions: carry or not. Asymmetric costs. Derived threshold.

If your paper can show your reader that your fraud threshold and your umbrella threshold are the same equation, your reader will understand your agent in one paragraph. Put the umbrella calculation early in the paper for exactly this reason.

Extend it if you want to go further: what if you could check the sky for free before leaving? That is a value-of-information calculation on a one-bit observation, and it is the same arithmetic as Section 14.

## 17 The generalization challenge

This is the most important thinking exercise of the week. It is what separates a design from a finding.

Take your agent and imagine it deployed somewhere different. A startup agent moves into an enterprise. A consumer tool moves into a hospital. An Indian deployment moves to Germany.

### 17.1 First, change the environment

Answer all eight:

1. Which assumptions break?
2. Which probabilities change, and in which direction?
3. Which costs change?
4. Which hidden states were missing in the original design?
5. Which evidence becomes more important?
6. Which evidence becomes less important, or unavailable?
7. Does the decision threshold stay the same?
8. Does the human-review policy stay the same?

Worked briefly for the running example, moving from a 20-person startup to a 5,000-person enterprise:

|                        | Startup                                                                   | Enterprise                                                                                     |
|------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Priors                 | Fraud is rare; few suppliers; the finance lead knows every one personally | Fraud attempts are constant; thousands of suppliers; nobody knows them                         |
| Best evidence          | “Does this feel like Ravi?” — personal familiarity does most of the work  | Personal familiarity is unavailable; procurement records and vendor-master history do the work |
| Missing hidden state   | —                                                                         | Insider collusion. Nobody models it at 20 people. It is real at 5,000                          |
| Cost of false positive | Low; call the supplier, it is fixed in ten minutes                        | High; ticket queues, SLA breaches, a three-day approval chain                                  |
| Threshold              | Can be aggressive                                                         | Must be looser, or the queue collapses                                                         |
| Human review           | The founder reviews everything                                            | Impossible. Review must be rationed by expected loss                                           |

Notice that the single most valuable evidence source in the startup — personal familiarity — does not exist in the enterprise, and the most dangerous hidden state in the enterprise does not exist in the startup. Neither of these is a tuning issue. Both require redesign.

## 17.2 Then change one variable at a time

For each of these, say what changes and why:

1. **The cost of a false negative becomes 100× higher.** Your threshold moves by roughly two orders of magnitude. Does your action set survive that? Do you now verify everything? Is verification capacity the real constraint?
2. **Obtaining evidence becomes 10× more expensive.** Which checks fall out of the policy? Does the agent shift from asking to escalating? Does the cheap-checks-first ordering change?
3. **Historical data becomes unreliable.** Your priors were learned from it. How would you even notice? (Section 7 gives you the JSD alarm.) What does the agent do while it is unsure whether its own priors are valid?

The habit being trained here is one question: *what happens if the world changes?* An agent that is correct only in the environment it was tuned in is not a finished system. Say clearly in your paper which parts of your design are portable and which parts are local.

## 18 The experiment

Extend your Week 1 experiment. You must compare at least two strategies.

### 18.1 Minimum acceptable experiment

- **Cases:** at least 100. Simulated is fine. Each case has a true hidden state that the agent cannot see, and evidence generated from your likelihood tables.
- **Policies:** at least two, plus a baseline. See below.
- **Metrics:** at least accuracy, total cost, number of questions asked, and human-review rate.
- **Output:** one table and one figure.

## 18.2 The policies

| Policy | Name                 | Behaviour                                                                                                                           |
|--------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| P0     | Baseline             | Always take the most common action. No evidence at all. This exists to prove your agent beats doing nothing.                        |
| P1     | Belief only          | Update on freely available evidence. Act on the highest-probability state. No cost model, no threshold.                             |
| P2     | Threshold            | Update on free evidence. Act by expected cost against a derived threshold. Escalate in the uncertain band.                          |
| P3     | Value of information | As P2, but actively chooses which paid check to run, ranked by expected information gain per unit cost, and stops by the stop rule. |

P0 and P3 are the interesting endpoints. If you only have time for three, run P0, P2, and P3.

### Do not skip P0

The baseline is what makes your result meaningful. In a domain where 96% of requests are genuine, an agent that approves everything is 96% accurate. If your sophisticated agent reports 94% accuracy, you need to know that the trivial policy beat it — and you need to explain why your agent is still better, which it may well be once cost is counted rather than accuracy.

This is a very common outcome in imbalanced problems and it is a good finding, not a failure.

## 18.3 Metrics, defined plainly

| Metric            | Plain meaning                                                                            |
|-------------------|------------------------------------------------------------------------------------------|
| Accuracy          | Fraction of decisions that were right. Nearly useless alone when classes are imbalanced. |
| Precision         | Of the requests the agent flagged, how many were actually bad.                           |
| Recall            | Of the actually bad requests, how many the agent caught.                                 |
| Decision cost     | Total money lost to wrong decisions. <i>Report this as your headline number.</i>         |
| Information cost  | Total spent on checks.                                                                   |
| Questions asked   | Average number of paid checks per case.                                                  |
| Human-review rate | Fraction escalated. Measures the load your agent puts on people.                         |
| Calibration error | Average gap between stated confidence and observed frequency.                            |

## 18.4 The result table you are aiming for

| Policy       | Acc. | Prec. | Recall | Decision cost | Info cost | Human % |
|--------------|------|-------|--------|---------------|-----------|---------|
| P0 baseline  | 0.96 | —     | 0.00   | Rs. 32,00,000 | Rs. 0     | 0       |
| P1 belief    | 0.94 | 0.41  | 0.62   | Rs. 12,10,000 | Rs. 0     | 0       |
| P2 threshold | 0.89 | 0.24  | 0.94   | Rs. 2,40,000  | Rs. 0     | 18      |
| P3 VOI       | 0.93 | 0.38  | 0.93   | Rs. 2,55,000  | Rs. 6,200 | 6       |

*Illustrative shape only. These are not results. Your numbers will differ and should.*

Read what that shape is saying, because it is the lesson of the whole week: **the policy with the best accuracy has the worst cost, and the policy with the worst accuracy has nearly the best cost.** P3 buys back accuracy and cuts human load to a third of P2's, for six thousand rupees of checks. If your

results tell a story like this, you have understood the material. If your results show one policy dominating on every metric, look for a bug.

### 18.5 Practical notes

Simulation is completely acceptable and is what most of you should do. Generate the true hidden state from your priors, then generate evidence from your likelihood tables, then let each policy decide. You wrote the likelihoods yourself, so your simulation cannot validate your likelihoods — it validates your *policy*, given those likelihoods. Say this in your limitations section. It is an important distinction and stating it clearly is a mark of a careful researcher.

Fix your random seed and report it. Run each policy on the identical set of cases; otherwise you are comparing policies and luck at the same time.

## 19 Failure analysis

Find at least five cases where your agent got it wrong. Study each one properly.

This section is worth more to your grade than your results section. Anyone can report numbers. Reading your own failures honestly is the harder skill and the rarer one.

### 19.1 Ten questions per failure

1. What did the agent believe?
2. What was actually true?
3. Which evidence did it use?
4. Which evidence did it ignore or never obtain?
5. Was the probability wrong?
6. Was the hidden-state list incomplete?
7. Was the threshold wrong?
8. Was the cost model wrong?
9. Should it have asked for more information?
10. Should a human have been involved?

### 19.2 Classify each failure

| Category                 | What it means                                                        |
|--------------------------|----------------------------------------------------------------------|
| Missing hidden state     | The truth was not on the agent's list. It could not have been right. |
| Bad prior                | The starting belief was wrong.                                       |
| Bad evidence             | The signal was corrupted or unavailable.                             |
| Misleading evidence      | The signal was correct but pointed the wrong way.                    |
| Poor calibration         | The confidence number was not trustworthy.                           |
| Wrong threshold          | The belief was right; the cut-off was wrong.                         |
| Wrong cost assumption    | The threshold was derived from wrong costs.                          |
| Distribution shift       | The world changed and the priors did not.                            |
| Insufficient information | The agent acted before it should have.                               |
| Wrong action policy      | Everything was right and the mapping to actions was wrong.           |

### The mark of a good failure analysis

A weak analysis says: “the agent misclassified case 47.”

A strong one says: “Case 47 was insider collusion. The agent held belief across four states, none of which was insider collusion, so the posterior was forced to distribute across three wrong explanations and it settled confidently on ‘genuine’ at 0.81. No evidence source could have fixed this, because every likelihood in my table was conditioned on states that did not include the truth. Category: missing hidden state. I added a fifth state with a 3% prior and a residual ‘other’ state at 1%, and re-ran. Cost fell by 18%, human-review rate rose by 4 points.”

The second version diagnoses, categorises, fixes, and re-measures. Aim for that on at least two of your five.

### 19.3 Do not hide bad results

If your clever policy lost to the baseline, report it and explain why. A paper that says “my information-theoretic policy underperformed the trivial rule, and here is the mechanism” is substantially better research than a paper with a suspiciously clean win. Negative results that are honestly explained are the most credible thing a student can submit.

## 20 Social learning

Continue the same workflow as Week 1. The purpose is not to become a content creator. It is to put your thinking in front of real people and find out what you assumed without noticing.

### 20.1 LinkedIn: five posts

Five posts, connected to your project. Not five generic AI posts. Suggested spread:

1. Your original problem and why it is genuinely hard.
2. One probabilistic insight from your own work.
3. One information-theoretic insight from your own work.
4. A surprise, a disagreement, or an experiment that did not go as expected.
5. Your preprint, or the single most important thing in it.

### The requirement that matters

At least one post must contain something **you actually discovered**, not something repeated from a class or from this document.

The template of a good post:

*“I assumed X. I tested it. X was wrong, and here is the number that showed it. What I now believe instead is Y.”*

A real example from this document’s running case: “I assumed the best question is the one that gives the most information. Then I priced the questions. My highest-information check costs Rs. 10,000 and buys 0.009 bits per hundred rupees; a database lookup costs Rs. 2 and buys 17. Once cost entered the model, the optimal question inverted completely.”

Five definitions posted as five posts will not meet this requirement. We are looking for evidence that your thinking changed.

## 20.2 Reddit and X

Continue what you started in Week 1. Ask technical questions. Reply to people. Look actively for disagreement, for assumptions you did not know you had, and for real-world failure cases from people who have operated systems like yours.

Do not repost your LinkedIn content. The formats reward different things and people notice.

The question to answer honestly this week:

**Did another human tell you something that changed your understanding of your problem?**

If yes, record what they said, what you believed before, and what you changed.

If no, say so, and explain why — wrong communities, questions too broad, not enough time, nobody engaged. “No” is an acceptable answer. Pretending otherwise is not.

## 20.3 The discussion record

**Update your existing discussion-record.md.** Do not create a separate Week 2 file. The point is a continuous record of how your understanding developed, and splitting it destroys that.

For each useful exchange: where it happened, what you asked, what they said, and — the column people forget — what you changed as a result. If the last column is empty for every row, the conversations were not useful and you should ask better questions.

## 21 The preprint

Your submission is one PDF, in IJCAI-style format, written by you.

### 21.1 Title

Specific to your problem. Not “Information Theory in AI Agents”.

Good: *A Probabilistic Decision Framework for Supplier Payment Verification Under Uncertainty*. The reader should know your problem, your method, and your setting from the title alone.

### 21.2 Section by section

| # | Section            | What it must contain                                                                                                                                                           |
|---|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Abstract           | Problem, approach, the probabilistic view, the information-theoretic extension, the experiment, and the main finding. Roughly 150–200 words. Write it last.                    |
| 2 | Introduction       | The real problem. Why uncertainty exists here. Why asking an LLM directly is not sufficient. Why the agent needs beliefs, why information selection matters, why cost matters. |
| 3 | Related Work       | What you read and <i>how it changed your design</i> . If you read nothing, write one honest paragraph saying so. That is better than fake citations.                           |
| 4 | Probabilistic View | Hidden states, priors and their origin, evidence, likelihoods, posteriors, one full worked update, the threshold.                                                              |

| #  | Section                    | What it must contain                                                                                                                                       |
|----|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5  | Information-Theoretic View | Only the concepts you actually used. Explain each in plain words plus one number from your problem. Do not include a concept because it sounds impressive. |
| 6  | Information Selection      | The evidence comparison table. Information value against information cost. Why the agent picks what it picks.                                              |
| 7  | Decision Policy            | Action set, derived threshold, expected cost, escalation rules, stop condition.                                                                            |
| 8  | Experiment                 | Data or simulation, number of cases, seed, baseline, policies, metrics, setup. Enough that someone could rebuild it.                                       |
| 9  | Results                    | At least one table and one figure. Show what changed between approaches.                                                                                   |
| 10 | Failure Analysis           | At least five failures, classified, with what you changed.                                                                                                 |
| 11 | Human Discussions          | What people told you and whether it changed the design.                                                                                                    |
| 12 | Limitations                | Honest. See below.                                                                                                                                         |
| 13 | New Questions              | Mandatory. The questions your work created.                                                                                                                |
| –  | AI-use statement           | Mandatory. Section 22.                                                                                                                                     |
| –  | References                 | Only things you read.                                                                                                                                      |

### 21.3 On limitations

Cover at minimum: data limitations, simulation limitations, which probabilities are assumptions rather than measurements, which costs are assumptions, hidden states you suspect are missing, where the design would not generalise, and what human oversight the system still requires.

A strong limitations section makes a paper more credible, not less. Reviewers trust authors who found their own weaknesses before the reviewer did.

### 21.4 On new questions

End with the questions your work generated. Not questions from this document — questions your own results raised. Examples of the shape:

- Can an agent learn which evidence is valuable without being told the likelihoods?
- How should an agent behave when two evidence sources reliably disagree?
- Can information value be estimated from a log of historical decisions?
- Should the decision threshold adapt per case, and what would it adapt on?
- How does an agent detect that its own hidden-state list is incomplete?

This section is where we look for research instinct. Three sharp questions that clearly come from your own results beat ten generic ones.

## 22 AI-use statement

Mandatory. A short, specific paragraph.

State which AI tools you used and for what. State which parts were AI-generated or AI-assisted. State how you verified the output. State which experiments you personally ran.

**Three rules**

1. Never claim an experiment happened if it did not.
2. Never cite a paper you did not read.
3. Never publish an AI-generated claim you cannot explain.

Heavy AI use, honestly declared, is fine and expected. This is an AI-native course. Undeclared AI use, or declared use that hides a fabricated result, is not.

An example of the right level of detail:

*I used Claude to find technical terms for supplier fraud, to explain KL divergence and Jensen–Shannon divergence in simple language, and to debug the simulation code. I wrote the hidden-state model, the likelihood tables, and the cost assumptions myself, based on two conversations with a finance operations lead (recorded in `discussion-record.md`). I ran the simulation myself; the results in Table 3 are from 500 simulated cases at seed 42. I verified the entropy and KL calculations by hand for the four-state case in Section 5 before trusting the code. The abstract was drafted with AI assistance and rewritten by me.*

**23 Repository and submission**

```
week2/
|-- README.md
|-- research-file.md
|-- discussion-record.md      <- updated, not replaced
|-- paper/
|   |-- main.tex
|   |-- references.bib
|   |-- figures/
|   |-- preprint.pdf         <- the deliverable
|-- src/
|-- notebooks/
|-- experiments/
|-- results/
|-- decisions/
|   |-- decision-record.md
|-- social/
|   |-- linkedin-posts.md
|   |-- x-thread.md
```

Everything except the PDF is optional. But:

**The PDF must stand alone.** A reader must be able to understand your entire project without opening your repository. If a result only makes sense with the notebook open, the paper is not finished.

**24 What we are evaluating**

Not the difficulty of your mathematics. This loop:

Observe → Question → Research → Hypothesize → Build → Test → Fail →  
Update → Generalize

A student who builds something complicated they do not understand has not met the objective.

A student who builds something simple, finds a real failure, explains why it failed, finds an idea that addresses it, changes the design, re-tests, and ends with a sharper question than they started with has done excellent work.

| Dimension           | Weight | What a strong submission looks like                                                         |
|---------------------|--------|---------------------------------------------------------------------------------------------|
| Problem formulation | 10%    | Same problem as Week 1, sharpened. Hidden states are genuinely hidden and genuinely matter. |
| Original reasoning  | 20%    | At least one conclusion that is yours, with the reasoning shown.                            |
| Research depth      | 15%    | Concepts found beyond what was taught, explained simply, used for something real.           |
| Experimentation     | 20%    | Two or more policies, a baseline, honest metrics, reproducible setup.                       |
| Failure analysis    | 15%    | Five failures, diagnosed and classified, with at least one design change and re-test.       |
| Generalization      | 10%    | Clear account of what breaks when the world changes.                                        |
| Technical writing   | 10%    | A reader outside your domain can follow it. Numbers labelled. Claims sourced.               |

The final standard is one question. Your PDF should answer it:

If I give an AI agent an uncertain world, how can it reason about what might be true, decide what information is worth obtaining, account for the cost of obtaining it, and eventually decide when it is rational to act?

You are not expected to have the final answer. You are expected to show that you can investigate the question scientifically.

## 25 Frequently asked questions and known ambiguities

This section exists because this brief is long and parts of it can be read more than one way. Read it when you are stuck.

### 25.1 Scope and effort

**How long should the paper be?** Eight to fourteen pages including tables and figures. A strong six-page paper beats a padded twenty-page one. We do not count pages.

**Do I have to answer all 98 questions in the appendices?** No. Answer the **20 required questions** in Appendix A inside your paper, in an appendix of your own. The rest are interview preparation. Work through them at your own pace, in your own notes.

**Do I need to implement every concept in Section 7?** No. Implement what your agent uses. Understand the rest well enough to explain simply. A concept in your paper that you cannot explain out loud is a liability, not a credential.

**Why does Section 7 have prompts instead of formulas?** Because a formula you cannot read teaches you nothing, and because going and learning the thing yourself is the actual assignment. Use the prompts, then write the explanation in your own words from memory. If you want the formulas, every prompt asks the assistant for them at the end — after you understand what they are for.

**Is this due in seven days?** No. See Section 3.

**I have read this whole brief and I still do not know where to start.** That is a normal reaction and it is not a sign you are behind. Run the readiness interview described in Section 2. It asks you about a dozen questions, accepts “I don’t know” as a useful answer, and produces a roadmap for your situation specifically.

**Do I have to use the skill files?** No. They are there because one document cannot give the right next step to a room full of people at different starting points.

## 25.2 Numbers, data, and honesty

**I have no real data. Can I make numbers up?** Yes, if you label them. Write “hypothetical” or “assumed” beside them and say where the assumption came from. Simulation with declared assumptions is a legitimate method. Undeclared invention is not.

**How many cases does the experiment need?** At least 100. Five hundred is comfortable for a simulation and costs nothing extra.

**My simulation uses the same likelihoods that my agent uses. Is that circular?** Partly, yes, and you should say so explicitly. Your experiment tests whether your *policy* is good given those likelihoods. It cannot test whether the likelihoods are right. State this in limitations. You may go further and deliberately mis-specify the simulator — generate cases from likelihoods slightly different from the agent’s — to test robustness. That is a strong optional extension.

**Must all my probabilities be exact?** No. They must be defensible and they must sum to one.

## 25.3 Concepts people get stuck on

**Likelihood versus probability.**  $P(\text{evidence} \mid \text{state})$  is a likelihood: “if the sender were an impersonator, how often would SPF fail?”  $P(\text{state} \mid \text{evidence})$  is a posterior: “given that SPF failed, how likely is an impersonator?” They are different numbers and confusing them is the most common error in this project. Bayes’ rule exists precisely to convert the first into the second.

**Why must the states sum to one?** Because the list is meant to be exhaustive and mutually exclusive. If they sum to less than one, some possibility exists that is not on your list — and if that is true, add a residual “other” state and give it real probability. Doing so is usually the single best improvement a Week 2 agent can make.

**Information gain and mutual information look identical.** They are the same quantity. See Section 7.

**Which log base?** Base 2, so the answers are in bits and you can talk about yes/no questions. Natural log gives nats. Nothing changes except the unit. Say which you used.

## 25.4 Design questions with no single answer

**How many hidden states should I have?** Between three and six, plus a residual “other”. Two is usually a hidden state model in disguise; ten is usually a taxonomy nobody can estimate priors for.

**What if two evidence sources disagree?** That is Advanced Question 18 and there is no single right answer. Bayes handles it automatically if the sources are conditionally independent — the likelihoods multiply and the stronger one wins. The interesting case is when they are *not* independent, or when one of them is unreliable in a way your model does not represent. Disagreement between sources is often the best possible escalation trigger, and saying so with reasoning is a complete answer.

**What if no evidence is reliable enough?** Then the agent should act on its default safe action and escalate. An agent that has no safe default is an incomplete design. Name yours.

**Can I use an LLM inside my agent?** Yes. Note that an LLM’s stated confidence is usually badly calibrated, so if you use it to produce probabilities, check the calibration or treat its output as one piece

of evidence with its own likelihood table rather than as a posterior.

## 25.5 Process

**Nobody replied to my Reddit post.** Record that. It is a genuine finding about where practitioners in your domain actually are. Try a different community, or try answering someone else's question first — reciprocity works.

**Can I change my problem?** Strongly discouraged. If you must, explain the reason in the paper. The point of continuity is to see your thinking deepen on one thing rather than restart on five.

**Something in this document contradicts something else, or is unclear.** Tell us. Ambiguity in the brief is our bug, not yours. Meanwhile, choose the reading that produces more honest research, do that, and note the choice in your paper.

## A Required core questions

Answer these twenty in your paper, in an appendix. Keep each answer to two or three sentences, in your own words, using your own problem wherever possible.

1. What is a hidden state, and what are yours?
2. Why is directly predicting an answer sometimes insufficient?
3. What is a belief distribution, and why must it sum to one?
4. What is the difference between prior and posterior?
5. What is likelihood, and how does it differ from probability?
6. What does it mean for an agent to be uncertain?
7. What is entropy, and why does it represent uncertainty?
8. What is a bit, in plain words?
9. What is information gain, and how does it relate to entropy?
10. What is conditional entropy?
11. What is mutual information, and how does it differ from correlation?
12. What is KL divergence, and why is it not a distance?
13. Why is KL divergence asymmetric? Give a small example.
14. What is Jensen–Shannon divergence, and why was it introduced?
15. What is calibration, and why does it matter for an agent?
16. What is expected cost, and how did you derive your threshold from it?
17. What is value of information? When is more information not worth it?
18. Can a question be highly informative and still useless? Give your own example.
19. What is distribution shift, and how would your agent detect it?
20. When should your agent stop searching and act? State the rule.

## B Full question bank

These are for your own preparation. They are not required in the paper. Work through them gradually — they are close to the questions you will be asked in interviews.

### B.1 Probability fundamentals

1. What is probability in simple words?
2. What is a random variable?
3. What is an event?
4. What is a sample space?
5. What is a probability distribution?

6. What is a prior probability?
7. What is a posterior probability?
8. What is conditional probability?
9. What does Bayes' theorem help us do?
10. What is the difference between prior and posterior probability?
11. What happens to our belief when new evidence arrives?
12. Why can two people have different prior beliefs about the same problem?
13. What is likelihood?
14. What is the difference between likelihood and probability?
15. Why must the probability of all possible hidden states sum to one?

## **B.2 Agent reasoning**

16. What is a hidden state?
17. Why does an AI agent need to reason about hidden states?
18. Why is directly predicting an answer sometimes insufficient?
19. What is a belief distribution?
20. Why should an agent maintain multiple possible explanations?
21. What does it mean for an agent to be uncertain?
22. Should an agent always act when it has an answer?
23. When should an agent ask for more information?
24. When should an agent involve a human?
25. What is a decision threshold?
26. Why can two agents with the same probability make different decisions?
27. What is the difference between prediction and decision?
28. Why does the cost of an error matter?

## **B.3 Information theory**

29. What is information?
30. What is uncertainty?
31. What is entropy?
32. Why does entropy represent uncertainty?
33. What happens to entropy when uncertainty decreases?
34. What is information gain?
35. Why does information gain relate to entropy?
36. What does a "bit" mean?

37. Why can eight equally likely possibilities require three bits?
38. Why is “Is it an animal?” sometimes better than “Is it a dog?”
39. What makes a question informative?
40. Can a question be informative but still useless?
41. Why does the agent’s objective matter when choosing information?
42. What is conditional entropy?
43. What is mutual information?
44. What does mutual information tell us intuitively?
45. How is mutual information different from correlation?
46. What is cross-entropy?
47. Why is cross-entropy commonly used in classification?
48. What is KL divergence?
49. Why is KL divergence useful?
50. Why is KL divergence not a true distance?
51. Why is KL divergence asymmetric?
52. What is Jensen–Shannon divergence?
53. Why might Jensen–Shannon divergence be preferred over KL divergence?
54. What does it mean when KL divergence is zero?
55. Can two distributions have low KL divergence but still represent different realities?

#### **B.4 Decision-making**

56. What is expected value?
57. What is expected cost?
58. Why should an agent consider cost?
59. Can the most probable action be the wrong action?
60. Why can a 35% probability still justify carrying an umbrella?
61. Why can the same probability produce different decisions in different situations?
62. What is opportunity cost?
63. What is value of information?
64. When is gathering more information not worth the cost?
65. Why should an agent sometimes stop searching?
66. What happens when the cost of being wrong becomes very large?

## B.5 Generalization

67. What happens if the probability model is wrong?
68. What happens if an important hidden state is missing?
69. What happens if the evidence is misleading?
70. What happens if historical data is not representative of the current situation?
71. What happens when the environment changes?
72. Why can a model be confident and still be wrong?
73. What is calibration?
74. Why is calibration important for an AI agent?
75. What is distribution shift?
76. How can an agent detect that its assumptions may no longer be valid?
77. When should an agent stop trusting historical evidence?
78. What should an agent do when two pieces of evidence disagree?

## C Advanced questions to investigate yourself

These are deliberately not fully answered anywhere in this document. Several are open. Pick the three most relevant to your problem and answer them properly in your paper — ideally with a number or an example from your own work rather than a definition.

1. If an agent has multiple possible hidden states, how should it represent uncertainty over them?
2. What happens when evidence is relevant to one hidden state but irrelevant to another?
3. Can adding more information ever make an agent's decision worse? Give an example.
4. Can more information increase uncertainty rather than decrease it? Explain with an example.
5. What is the difference between “I have more information” and “I have more useful information”?
6. How does KL divergence help compare two probability distributions?
7. Why is KL divergence asymmetric? Construct a simple example.
8. Why was Jensen–Shannon divergence introduced?
9. What is mutual information actually measuring?
10. Can two variables have strong dependence without ordinary correlation capturing it well?
11. What happens when the agent's probability estimates are poorly calibrated?
12. Why might a highly confident agent be more dangerous than an uncertain agent?
13. How can an agent decide whether to ask a question or act directly?
14. What is the relationship between information gain and decision-making?
15. Is maximizing information gain always the optimal strategy? Show why or why not with an example.

16. How does the cost of information change the optimal question?
17. What happens if the information itself can be unreliable?
18. How should an agent reason when two sources provide conflicting evidence?
19. How should an agent behave when no available evidence is sufficiently reliable?
20. What happens when the possible-world list itself is incomplete?

Questions 3, 4, 15, and 20 have worked answers hidden inside Sections 7, 12, and 14 of this document. Find them, then push further than this document does.

## D Submission checklist

### Social learning

- ☐ Five LinkedIn posts published, connected to the project.
- ☐ At least one post contains something you genuinely discovered.
- ☐ Reddit discussions continued from Week 1.
- ☐ X discussions continued from Week 1.
- ☐ `discussion-record.md` updated, not replaced.
- ☐ At least one idea identified that changed because of external feedback — or an honest explanation of why none did.

### Agent design

- ☐ Same problem as Week 1.
- ☐ Hidden states defined, including a residual state.
- ☐ Priors defined, with their origin stated.
- ☐ At least one full Bayes update shown by hand.
- ☐ Entropy reported before and after evidence.
- ☐ At least three evidence sources compared on value and cost.
- ☐ Five candidate questions tabled.
- ☐ Decision threshold derived from costs, not assumed.
- ☐ Expected-cost calculation shown.
- ☐ Human escalation rules stated.
- ☐ Stop-searching condition stated.

### Research

- ☐ At least two concepts learned beyond what was taught, explained simply and used.
- ☐ KL divergence investigated.
- ☐ Jensen–Shannon divergence investigated.

- ☐ Mutual information investigated.
- ☐ Conditional entropy investigated.
- ☐ Calibration investigated.
- ☐ Value of information investigated.
- ☐ Umbrella problem worked with your own cost assumptions.
- ☐ Generalization challenge answered for all three variable changes.

**Experiment**

- ☐ At least 100 cases.
- ☐ Baseline included.
- ☐ At least two policies compared.
- ☐ Seed reported; all policies run on identical cases.
- ☐ At least one table and one figure.
- ☐ At least five failures analysed and classified.
- ☐ At least one design change made because of a failure, and re-tested.

**Paper**

- ☐ One PDF, IJCAI-style structure.
- ☐ Specific title.
- ☐ Understandable without opening the repository.
- ☐ Hypothetical numbers labelled as hypothetical.
- ☐ Limitations honest and specific.
- ☐ New questions section included.
- ☐ Twenty required questions answered in an appendix.
- ☐ AI-use statement included.
- ☐ Every reference is something you actually read.

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*Ambiguity in this brief is our problem, not yours. If something here can be read two ways, tell us, and choose the reading that produces more honest research.*